

Jiacong Xu

Ph.D. Candidate, Department of Computer Science
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EDUCATION

Johns Hopkins University <i>Ph.D. in Computer Science</i> Advisor: Prof. Vishal M. Patel	Jan. 2023 – Present
Texas A&M University <i>M.S. in Electrical Engineering</i> Advisor: Prof. Shankar P. Bhattacharyya	Sep. 2019 – Dec. 2021
University of Science and Technology of China <i>B.Eng. in Mechanical Engineering</i>	Sep. 2015 – May 2019

RESEARCH INTERESTS

World Models | Video Generation | 3D Reconstruction | Vision-Language Models

RESEARCH EXPERIENCE

Meta Superintelligence Labs <i>Research Scientist Intern</i> Streaming and real-time video generation	Jun. 2026 – Present
Adobe Research <i>Research Scientist Intern</i> Video world models	Jan. 2026 – May 2026
Honda Research Institute USA, Inc. <i>Research Intern</i> Multimodal large language models	May 2024 – Aug. 2024

SELECTED PUBLICATIONS

- Wonder: Video World Model Done Better**
Jiacong Xu*, Hanwen Jiang, Zhixin Shu, Kalyan Sunkavalli, Vishal M. Patel, Yiqun Mei*
Technical Report, 2026
- InstantHDR: Single-forward Gaussian Splatting for High Dynamic Range 3D Reconstruction**
Dingqiang Ye*, Jiacong Xu*, Jianglu Ping, Yuxiang Guo, Chao Fan, Vishal M. Patel
ECCV 2026
- Endless World: Real-Time 3D-Aware Long Video Generation**
Ke Zhang, Jiacong Xu, Yiqun Mei, Vishal M. Patel
CVPR 2026
- FreeViS: Training-free Video Stylization with Inconsistent References**
Jiacong Xu, Yiqun Mei, Ke Zhang, Vishal M. Patel
ICLR 2026
- Think Before You Diffuse: LLMs-Guided Physics-Aware Video Generation**
Ke Zhang, Cihan Xiao, Jiacong Xu, Yiqun Mei, Vishal M. Patel
Under Review, 2025

SELECTED PUBLICATIONS (CONTINUED)

6. **Towards Zero-Shot Anomaly Detection and Reasoning with Multimodal Large Language Models**
Jiacong Xu, Shao-Yuan Lo, Bardia Safaei, Vishal M. Patel, Isht Dwivedi
CVPR 2025 Highlight
7. **Filter Images First, Generate Instructions Later: Pre-Instruction Data Selection for Visual Instruction Tuning**
Bardia Safaei, Faizan Siddiqui, Jiacong Xu, Vishal M. Patel, Shao-Yuan Lo
CVPR 2025 Highlight
8. **XYScanNet: A State Space Model for Single Image Deblurring**
Hanzhou Liu, Chengkai Liu, Jiacong Xu, Peng Jiang, Mi Lu
CVPRW 2025
9. **Wild-GS: Real-Time Novel View Synthesis from Unconstrained Photo Collections**
Jiacong Xu, Yiqun Mei, Vishal M. Patel
NeurIPS 2024
10. **Reference-based Controllable Scene Stylization with Gaussian Splatting**
Yiqun Mei*, Jiacong Xu*, Vishal M. Patel
NeurIPS 2024
11. **Leveraging Thermal Modality to Enhance Reconstruction in Low-Light Conditions**
Jiacong Xu, Mingqian Liao, Ram Prabhakar Kathirvel, Vishal M. Patel
ECCV 2024
12. **Animal3D: A Comprehensive Dataset of 3D Animal Pose and Shape**
Jiacong Xu, Yi Zhang, et al., Alan Yuille, Adam Kortylewski
ICCV 2023
13. **PIDNet: A Real-time Semantic Segmentation Network Inspired by PID Controllers**
Jiacong Xu, Zixiang Xiong, Shankar P. Bhattacharyya
CVPR 2023
14. **Communication-Efficient Design of Learning System for Energy Demand Forecasting of Electrical Vehicles**
Jiacong Xu, Riley Kilfoyle, Zixiang Xiong, Ligang Lu
APSIPA ASC 2023

* Equal contribution. Publication titles link to the corresponding papers.

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB, JavaScript, Verilog HDL, LabVIEW
Languages	Mandarin (native), English (fluent)

SELECTED COURSEWORK

Computer Vision	Computer Graphics	Natural Language Processing	Deep Learning
Mathematical Statistics	Stochastic Processes	Mathematical Physics	Complex Analysis